Lab 1 Outline

GradPace - Team Copper

Old Dominion University

CS 410 Professional Workforce Development I - Spring 2026

Professor Sarah Hosni

April 3, 2026

1. Introduction

- Workforce expectations are rapidly evolving
- 63% of employers cite skill gaps as top barrier to growth [World Economic Forum, 2025]
- 66% of enterprises are reducing entry-level hiring due to AI [Laurent, 2026]
- 91% report automation is changing or eliminating jobs [Laurent, 2026]
- 50-60% of recent graduates are underemployed [Cengage Group, 2025]
- Problem exists in higher education, career services, and entry-level job markets
- 48% of recent graduates feel underqualified for entry-level roles [Cengage Group, 2025]
- [Figure 1: Fluctuations in perception of preparedness in recent graduates]
- Academic progress is tracked, but career preparation is fragmented
- Current tools address isolated needs only
    - Coursework tracking (students can track their academic data, but not professional information)
    - Job boards
    - Networking platforms
- Gap: no unified, real-time career readiness roadmap
- Key problem characteristics
    - Fragmented data career development systems
    - Delayed gap detection
    - Reactive career decision-making
    - Limited visibility into career readiness
- [Figure 2: Educator POV on teaching responsibilities]
- Introduce GradPace as a unified, career readiness platform

2. Product Description

- GradPace: Career readiness platform designed to bridge academic progress and professional employment
- Built by Team Copper for students, alumni, and industry professionals
- Focuses on proactive preparation instead of traditional job boards
- Student Input and Roadmap
    - Students input academic and professional information
    - System generates a personalized roadmap
    - Aligns current progress with recommendations toward real industry standards
- Core Capabilities
    - Early Warning Gap Detection
    - Industry Readiness Dashboard
- Key Functions
    - Identifies what a student needs before entering the job market
    - Integrates live industry data
    - Includes GPS-based local networking
- Outcome
    - Users are not just tracking past achievements
    - Users are synchronized with modern workforce requirements

2.1. Key Product Features and Capabilities

- **Account Management:** Includes Login authentication, Account Creation, and Account Subscription Management.
- **Personal Academic and Professional Info Input:** An interface for students to enter their specific background towards goals.
- **Personalized Roadmap Generation:** A tool that builds a custom roadmap for the student based on their data.
- **View Roadmap:** A visual dashboard where students can see and track their progress.
- **Early-Warning Gap Detection:** One of GradPace's most unique capabilities is the ability to catch "gaps" in a student's resume or skills early enough to bridge their goals.
- **Industry Readiness Dashboard:** The system can evaluate a user's standing against real-world standards to see if they are ready for a job.
- **Location-Based Event Recommendations:** The platform can identify and suggest professional events based on the user's physical location.
- **Alumni Survey:** GradPace allows alumni to complete a survey about current industry trends related to things like desired skills. This survey also provides alumni an opportunity to earn some of the money back that they spent on a GradPace subscription.
- **Industry Data:** GradPace stays relevant by constantly taking industry data from actual industry professionals, ensuring students aren't following outdated advice.

2.2. Major Components (Hardware/Software)

- GradPace is a web-based platform built around four main software components
    - Frontend
    - Backend
    - Database
    - API Integration
- Frontend hosts UI (account management, information tracking, dashboard)
- Backend manages system logic (security, roadmap, gap detection systems, benchmark comparisons, and development recommendations)
- Data later stores structured info like user profiles, records, user skills, and opportunity listings through PostgreSQL
- GradPace connects to live industry data, location services, and AI systems for recommendations and roadmap refinement.
- Users need internet connection and other generally standard computer systems to access our site, but all other data is hosted in the cloud.

3. Identification of Case Study

- **WHO:** Alex, a computer science undergraduate student in his second semester of college.
- **WHAT PROBLEM:** Unsure how to begin professional computer science development outside of general classwork.

- **FEATURES:** Opportunity finding, custom roadmap
- **SCENARIO:** Alex inputs transcripts, current work interests, and location information, and now has access to a step-by-step roadmap to help develop his skills to prepare for that post-grad goal
- **FUTURE:** University integration, alumni engagement, market trend data use to keep university courses relevant

4. Glossary

API (Application Programming Interface): A set of rules that allow different software applications to communicate and exchange data with each other.

Authentication: The process of verifying a user's identity before allowing access to a system.

Backend: The server-side part of a website or application that handles logic, databases, and data processing.

Career Readiness: A measure of how prepared a student is to enter a specific career based on skills, experience, and qualifications.

Career Roadmap: A structured, visual plan that outlines a student's progress, goals, and required steps toward a specific career path.

Cloud Computing: The use of remote servers on the internet to store data and run applications instead of a local computer.

Data Leak: The accidental or unauthorized exposure of sensitive or private information.

Database: An organized collection of data that can be stored, accessed, and managed electronically.

Encryption: A security method that converts data into a coded form so that only authorized parties can read it.

Frontend: The user-facing part of a website or application that users interact with in their browser.

Gap Detection (Early-Warning System): A feature that identifies missing skills, experiences, or qualifications needed for a target career.

Industry Readiness Benchmarking: The process of comparing a student's profile against expected skills and qualifications for specific careers.

Location-Based Event Recommendations: Suggestions for nearby internships, networking events, or career opportunities based on user location.

Professional Data Input: User-provided information such as coursework, projects, internships, and certifications used to build a profile.

Survey Results Dashboard: A feature that displays aggregated insights from alumni data to inform students and administrators.

System Administrator: A role responsible for managing the platform, maintaining data quality, and overseeing system functionality.

Token (AI): A small unit of text processed by an AI model, such as a word, part of a word, or punctuation.

User Interface (UI): The visual elements of a software application that allow users to interact with it.

Web Server: A computer system or software that stores websites and delivers web pages to users when they request them through a browser.

5. References

Brown, E. (2025, January 21). *New Survey Reveals Traditional Undergraduate Education Is Not Preparing Students for the Workforce*. Hult International Business School. https://www.hult.edu/blog/wi_skills_survey/

Cengage Group. (2025). *The Missing Link: Accountability in Career Readiness.* Cengage Group. https://www.cengagegroup.com/edtech-research/employability-report/

Laurent, A. (2026, March 8). *AI's Impact on Graduate Jobs: A 2025 Data Analysis.* IntuitionLabs. https://intuitionlabs.ai/articles/ai-impact-graduate-jobs-2025

Mardis, M. A., Ma, J., Jones, F. R., Ambavarapu, C. R., Kelleher, H. M., Spears, L. I., & McClure, C. R. (2018). Assessing Alignment Between Information Technology Educational Opportunities, Professional Requirements, and Industry Demands. *Education and Information Technologies, 23*(4), 1547–1584. https://doi.org/10.1007/s10639-017-9678-y

World Economic Forum. (2025, January 7). *Future of Jobs Report 2025*. World Economic Forum. https://www.weforum.org/reports/the-future-of-jobs-report-2025/